Artificial Intelligence Technology and Application

Machine Learning Lab Guide

Student Version

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1 Survival Prediction of the Titanic

1.1 Introduction

1.1.1 About This Lab

This experiment is to predict whether passengers on the Titanic can survive based on the Titanic datasets.

1.1.2 Objectives

Upon completion of this task, you will be able to:

- Use the Titanic datasets open to the Internet as the model input data.
- Build, train, and evaluate machine learning models
- Understand the overall process of building a machine learning model.

1.1.3 Datasets and Frameworks

This experiment is based on **train.csv** and **test.csv**. **test.csv** contains the result about whether the passengers can survive. This dataset has no target, that is, no result, and can be used as a real-world dataset. Involved parameters are as follows:

- **PassengerId**: passenger ID
- **Pclass**: cabin class (class 1/2/3)
- **Name**: passenger name
- **Sex**: gender
- **Age**: age
- **SibSp**: number of siblings/number of spouses
- **Parch**: number of parents/number of children
- **Ticket**: ticket No.
- **Fare**: ticket price
- **Cabin**: cabin No.
- **Embarked**: port of boarding

1.2 Procedure

1.2.1 Importing Related Libraries

```
import pandas as pd
import numpy as np
import random as rnd

import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline

from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC, LinearSVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.linear_model import Perceptron
from sklearn.linear_model import SGDClassifier
from sklearn.tree import DecisionTreeClassifier
```

1.2.2 Importing Datasets

Step 1 Read data.

Step 2 View data.

```
['PassengerId' 'Survived' 'Pclass' 'Name' 'Sex' 'Age' 'SibSp' 'Parch'
 'Ticket' 'Fare' 'Cabin' 'Embarked']
```

The first five rows of data are displayed.

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

The last five rows of data are displayed.

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	Q

The data overview helps check whether some data is missing and what the data type is.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 11 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  418 non-null    int64
 1   Pclass       418 non-null    int64
 2   Name         418 non-null    object
 3   Sex          418 non-null    object
 4   Age          332 non-null    float64
 5   SibSp        418 non-null    int64
 6   Parch        418 non-null    int64
 7   Ticket       418 non-null    object
 8   Fare         417 non-null    float64
 9   Cabin        91 non-null     object
 10  Embarked     418 non-null    object
dtypes: float64(2), int64(4), object(5)
memory usage: 36.0+ KB
```

The related numeric-type information of the data helps check the average value and other statistics.

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

The character-type information helps check the number of types, the type with the maximum value, and the frequency.

	Name	Sex	Ticket	Cabin	Embarked
count	891	891	891	204	889
unique	891	2	681	147	3
top	Vovk, Mr. Janko	male	CA. 2343	B96 B98	S
freq	1	577	7	4	644

Step 3 Check the survival probability corresponding to each feature based on statistics.

	Pclass	Survived
0	1	0.629630
1	2	0.472826
2	3	0.242363

The intuitive data shows that passengers in class 1 cabins are more likely to survive.

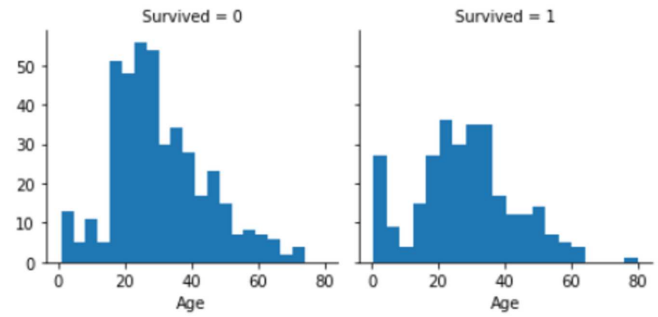
	SibSp	Survived
1	1	0.535885
2	2	0.464286
0	0	0.345395
3	3	0.250000
4	4	0.166667
5	5	0.000000
6	8	0.000000

The survival probability can be directly determined by the number of siblings.

	Sex	Survived
0	female	0.742038
1	male	0.188908

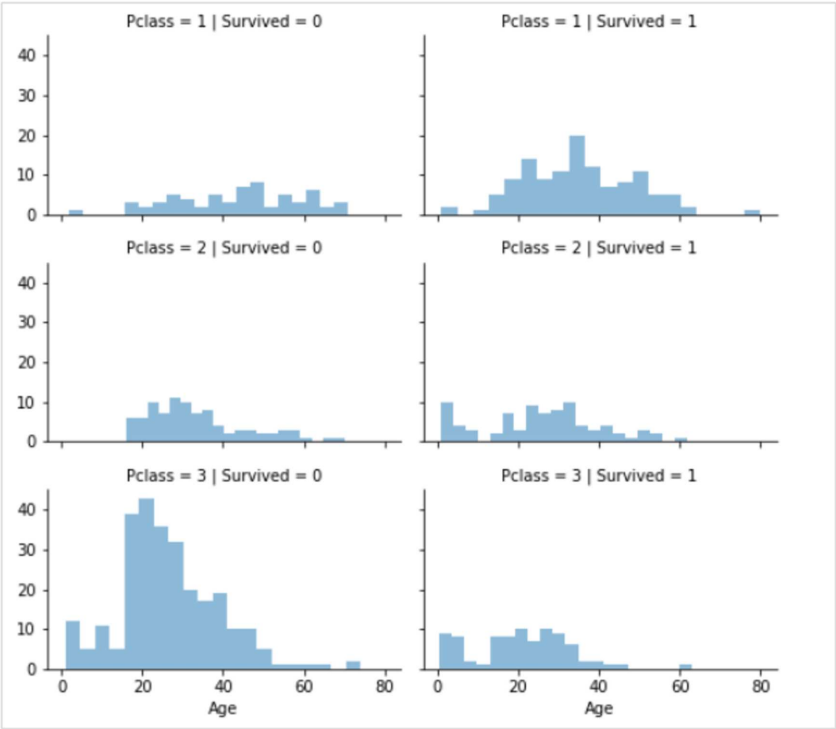
When the survival probability is determined by gender, an obvious imbalance occurs.

```
<seaborn.axisgrid.FacetGrid at 0x1a1df099a88>
```



As shown in the preceding figure, most young passengers died.

The following figure shows the survival probability determined based on the cabin and age.



----End

1.2.3 Preprocessing Data

As the datasets have missing values, combine the datasets, and fill the missing values with data.

Step 1 Combine the datasets.

Step 2 Check for missing values.

```
PassengerId       0
Survived        418
Pclass            0
Name              0
Sex               0
Age             263
SibSp             0
Parch             0
Ticket            0
Fare              1
Cabin          1014
Embarked          2
dtype: int64
```

Step 3 Fill the missing values with data.

Process the datasets by using different methods as required. For example, fill the **Fare** and **Embarked** parameters having few missing values with the mode.

Use the average age value.

Delete less significant data. Before this, assign a value to **Target** first.

Check whether missing values still exist.

Step 4 Convert data.

Convert some character-type data into numeric-type data for model input. To do so, check the number of types first.

```
male      843
female    466
Name: Sex, dtype: int64
```

Use the search function to obtain each character-type value and replace it with a numeric-type value.

test.csv cannot be used as a training test set as it does not contain **Target**. **train.csv** contains 891 pieces of data (with **Target**), which need to be extracted.

----End

1.2.4 Building a Model

This section describes how to build a model. To build a model, split the training set and test set.

Step 1 Split the dataset.

```
from sklearn.model_selection import train_test_split
```

Step 2 Train a model.

The logistic regression algorithm, random forest algorithm, and AdaBoost are used for training.

```
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn import ensemble
```

```
logR 0.7488789237668162
RFC 0.8071748878923767
AdaBoost 0.7757847533632287
```

As shown above, the random forest algorithm has a good effect.

Step 3 Predict data.

```
array([0., 1., 0., 0., 1., 0., 1., 0., 1., 0., 0., 0., 1., 0., 1., 1., 0.,
       0., 1., 1., 1., 0., 1., 1., 1., 0., 1., 0., 0., 0., 0., 0., 1., 1.,
       1., 0., 1., 1., 0., 0., 0., 1., 0., 1., 1., 0., 1., 0., 1., 1., 1.,
       0., 1., 1., 1., 0., 0., 0., 0., 1., 0., 0., 0., 1., 1., 1., 1., 0.,
       1., 1., 1., 0., 1., 1., 1., 1., 0., 1., 0., 1., 1., 0., 0., 0., 0.,
       0., 1., 1., 1., 1., 1., 0., 1., 0., 1., 0., 0., 0., 1., 0., 1., 0.,
       0., 0., 1., 0., 0., 0., 0., 0., 0., 1., 1., 1., 1., 0., 0., 1., 1.,
       1., 1., 0., 1., 0., 0., 1., 0., 1., 0., 0., 0., 1., 0., 0., 0., 0.,
       0., 0., 1., 0., 0., 1., 1., 0., 0., 0., 1., 0., 1., 0., 1., 0., 0.,
       1., 0., 0., 1., 1., 0., 1., 1., 0., 1., 0., 0., 1., 1., 0., 1., 1.,
       0., 0., 0., 0., 0., 1., 1., 1., 1., 0., 0., 1., 1., 0., 1., 0., 1.,
       0., 0., 0., 0., 1., 0., 0., 1., 0., 1., 1., 0., 1., 1., 1., 1., 1.,
       0., 1., 1., 0., 1., 0., 0., 0., 1., 1., 0., 0., 1., 0., 1., 0., 1.,
       0., 1., 0., 1., 1., 0., 1., 0., 0., 1., 1., 0., 0., 1., 0., 0., 0.,
       1., 1., 1., 1., 1., 0., 0., 0., 1., 0., 1., 1., 1., 0., 1., 0., 0.,
       0., 0., 0., 1., 0., 0., 0., 1., 1., 0., 0., 0., 0., 1., 0., 1., 0.,
       1., 1., 0., 1., 0., 0., 0., 0., 1., 1., 1., 1., 1., 0., 0., 1., 0.,
       0., 1., 1., 0., 1., 0., 0., 1., 0., 1., 0., 0., 0., 0., 0., 1., 0.,
       0., 1., 0., 1., 0., 0., 0., 1., 1., 1., 0., 0., 0., 0., 0., 0., 0.,
       1., 1., 0., 1., 1., 0., 0., 1., 1., 0., 1., 0., 1., 0., 0., 0., 0.,
       0., 0., 0., 1., 0., 1., 0., 1., 0., 1., 1., 0., 1., 0., 1., 0., 1.,
       0., 0., 1., 0., 1., 1., 0., 1., 1., 0., 1., 1., 0., 0., 1., 0., 0.,
       1., 1., 1., 0., 0., 0., 0., 0., 1., 1., 0., 1., 0., 0., 0., 0., 1.,
       1., 0., 0., 0., 1., 0., 1., 0., 0., 1., 0., 1., 1., 1., 0., 0., 1.,
       1., 1., 1., 1., 1., 0., 1., 0., 0., 0.])
```

----End